8-4 Mechanism of Web Pages and Emails

Point!

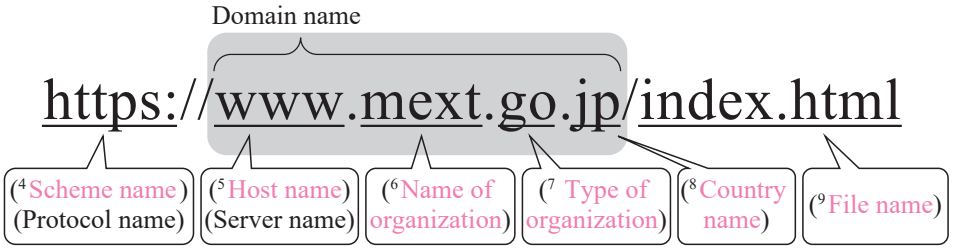
1 Mechanism of the Internet

(1) ([1]**WWW**) : A system that links countless pieces of content on the internet. Users can browse web pages that compile various data. It is commonly referred to as the Web.

(2) ([2]**HTML**) : A language used for creating web pages.

(3) ([3]**URL**) : Represents the location of a web page and the method of communication.

<Structure of a URL>



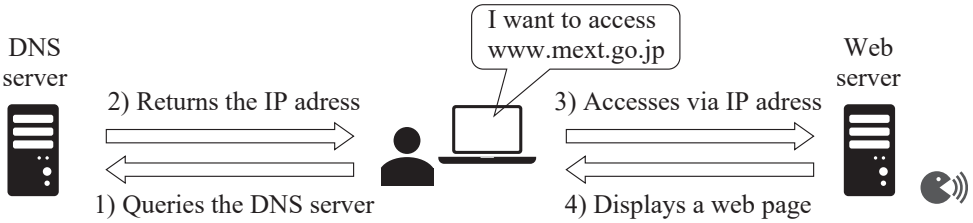
<Examples of Types of Organizations>

Type	Meaning
gov	Government agency
co	Corporation
edu	School (elementary, junior high, and high schools)
ac	University/educational corporation

<Examples of Country Codes>

Code	Country name
eg	Egypt
jp	Japan
hk	Hong Kong
kr	South Korea

(4) Process of Browsing Web Pages



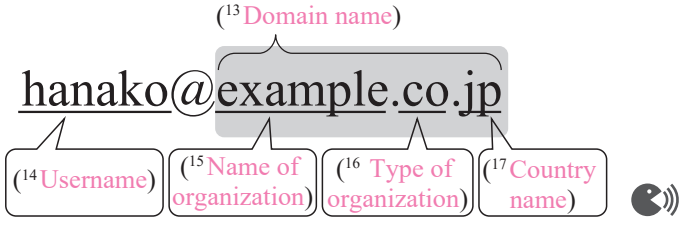
2 Mechanism of Email

(1) Email : A system for exchanging messages and data with specific users via the internet. When sending an email, you use an ([10]**email address**).

(2) Structure of an Email Address

[1] ([11]**Username**) : Identifies the recipient.

[2] ([12]**Domain name**) : Identifies the mail server that the recipient is using.

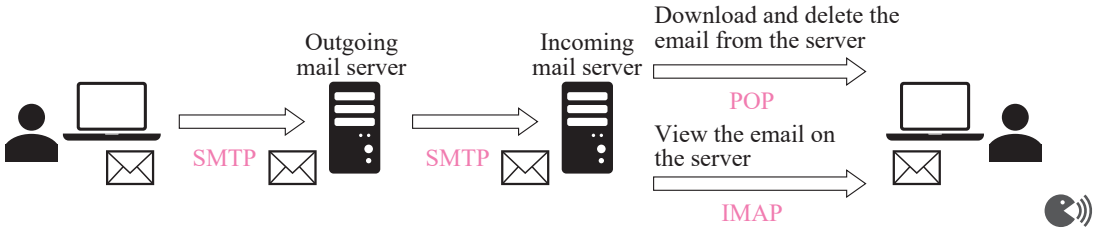


(2) Process of Sending and Receiving Emails

[1] ([18]**SMTP**) : A protocol for sending emails to a mail server.

[2] ([19]**POP**) : A protocol used to download emails from a server and delete them from a server.

[3] ([20]**IMAP**) : A protocol for accessing emails stored on a mail server.



Warm Up

Answer the following questions.

(1) You entered "https://www.example.co.jp" in the web browser and displayed the web page. Rearrange the steps **A** to **D** in the order of processing to display the web page.

A The web server sends the data of the web page.

B Access the IP address returned from the DNS server.

C The DNS server returns the IP address of the web server corresponding to the domain name.

D Query the DNS server for the IP address corresponding to the domain name.

(2) Fill in the blanks [1] to [4] in the following sentence with the appropriate terms.

> An email address is composed of "username@([1])," and the protocol ([2]) is used for sending emails, while ([3]) or ([4]) is utilized for receiving emails.

Explanation

(1) **D → C → B → A**

(2) [1] Domain name [2] SMTP [3] POP [4] IMAP (note: [3] and [4] can be in any order)

Try

1 Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [4] in the following sentences from the options **A** to **E** below, and answer using the letters.

> Web pages are written in a language called ([1]) and the location and communication protocol of the web page are specified by ([2]). The system for converting domain names and IP addresses within ([2]) is ([3]), and ([4]) fulfills that role.

A URL **B** DNS **C** HTML
D WWW **E** DNS server

(2) What is the name of the service that allows you to view web pages on the internet?

(3) Choose the appropriate names for sections [1] to [6] that compose the following URL from the options **A** to **G** below, and answer using the letters.

https://www.mext.go.jp/index.html
[1] [2] [3] [4] [5] [6]

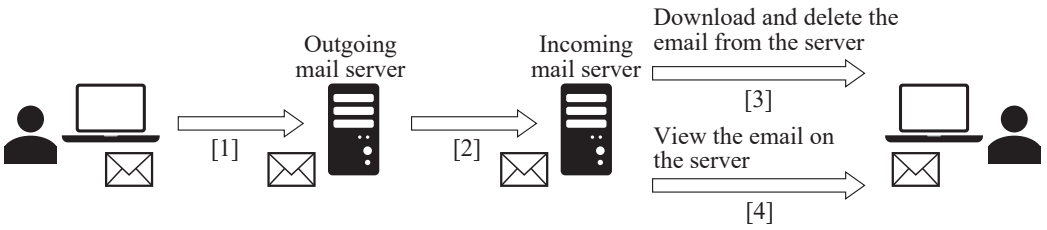
A Scheme name **B** Country name **C** Organization type name **D** Name of organization
E Hostname **F** File name **G** Domain name

2 Answer the following questions.

(1) Name the components labeled [1] and [2] that constitute the following email address.

hanako@example.ed.jp
[1] [2]

(2) The following diagram represents the flow of sending and receiving emails. Choose the appropriate combination of protocols that fit in [1] to [4] in the following diagram from the options **A** to **D** below.



	[1]	[2]	[3]	[4]
A	POP	SMTP	SMTP	IMAP
B	POP	POP	IMAP	SMTP
C	SMTP	SMTP	POP	IMAP
D	SMTP	SMTP	IMAP	POP

Exercise

1 Answer the following questions.

(1) Choose the most appropriate explanation of a DNS from the options **A** to **D** below, and answer using the letter.

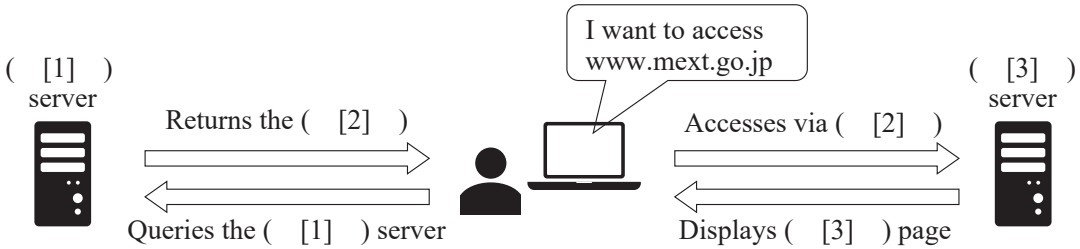
A A system that links countless pieces of content on the internet and is commonly called “the Web.”

B Represents the location of a web page on the internet and the method of communication.

C A language developed for creating web pages.

D A system that maps domain names to IP addresses and vice versa.

(2) The following diagram represents the flow of browsing a web page. Choose the term that best fits into the blanks [1] to [3] in the diagram from the options **A** to **D** below, and answer using the letters.



A Web **B** URL **C** IP address **D** DNS

(3) Choose the appropriate combination of names for the blanks [1] to [6] that compose the following URL from the options **A** to **D** below, and answer using the letter.

https://www.mext.go.jp/index.html
[1] [2] [3] [4] [5] [6]

	[1]	[2]	[3]	[4]	[5]	[6]
A	Scheme name	Hostname	Type of organization	Name of organization	Country name	File name
B	Scheme name	Hostname	Name of organization	Type of organization	Country name	File name
C	Hostname	Scheme name	Type of organization	Name of organization	Country name	File name
D	Hostname	File name	Name of organization	Type of organization	Country name	Scheme name

2 Choose the appropriate options for protocols [1] to [3] from the options **A** to **D** below, and answer using the letters.

(1) Protocol for sending emails to a mail server.

(2) Protocol for accessing emails on a mail server.

(3) A protocol that downloads emails from the mail server and deletes them from the server.

A POP **B** SMTP **C** DNS **D** IMAP